

CHARKOV, Russian Federation

WMO: 298640

Lat: 53.717N

Lon: 90.367E

Elev: 1724

StdP: 13.80

Time Zone: 7.00 (E07)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -26.9	(c) -22.3	(d) -31.9	(e) 1.0	(f) -26.7	(g) -27.4	(h) 1.3	(i) -22.0	(j) 21.1	(k) 21.8	(l) 18.7	(m) 18.0	(n) 5.5	(o) 320	(p) 0.635

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.5	83.6	63.9	80.6	62.8	77.2	61.5	66.4	78.6	64.7	76.0	63.2	73.9	5.8	110

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
62.2	89.1	71.2	60.4	83.5	70.0	58.7	78.6	68.5	32.1	78.6	30.7	76.1	29.6	73.9	76.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 19.3	(b) 16.5	(c) 13.9	(e) -30.7	(f) 89.6	(g) 6.5	(h) 3.6	(i) -35.3	(j) 92.1	(k) -39.1	(l) 94.2	(m) -42.7	(n) 96.2	(o) -47.4	(p) 98.8		
(a) 19.3	(b) 16.5	(c) 13.9	(e) -30.8	(f) 70.2	(g) 6.2	(h) 2.9	(i) -35.3	(j) 72.3	(k) -38.9	(l) 74.0	(m) -42.4	(n) 75.6	(o) -46.9	(p) 77.7		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	34.0	-0.6	5.9	22.2	37.6	49.2	62.0	64.3	59.9	47.6	33.8	18.4	5.6		
	DBStd	24.43	12.59	12.57	11.97	9.22	8.81	6.91	5.04	6.15	7.48	9.15	12.34	12.77		
	HDD50	7143	1568	1236	862	382	124	5	0	4	132	506	948	1376		
	HDD65	11450	2033	1656	1327	823	493	133	74	180	524	968	1398	1841		
	CDD50	1289	0	0	0	8	100	364	444	311	59	3	0	0		
	CDD65	123	0	0	0	0	3	43	53	23	1	0	0	0		
	CDH74	1448	0	0	0	2	81	575	478	286	26	0	0	0		
	CDH80	317	0	0	0	0	11	139	99	64	4	0	0	0		
	WSAvg	6.0	5.3	5.5	6.6	7.2	7.7	6.6	5.5	5.5	5.8	5.4	5.1	5.5		
	PrecAvg	11.7	0.2	0.1	0.2	0.5	1.1	2.4	2.8	2.3	1.1	0.4	0.4	0.2		
(15) Precipitation	PrecMax	18.8	0.5	0.4	1.1	1.6	4.4	5.1	4.3	4.8	2.6	1.9	1.0	0.5		
	PrecMin	7.2	0.0	0.0	0.0	0.0	0.2	0.6	1.5	0.6	0.2	0.0	0.1	0.0		
	PrecStd	2.4	0.1	0.1	0.2	0.4	0.9	1.1	0.7	1.2	0.6	0.4	0.2	0.1		
	PrecStd	2.4	0.1	0.1	0.2	0.4	0.9	1.1	0.7	1.2	0.6	0.4	0.2	0.1		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	36.7	37.2	55.6	69.9	81.8	88.5	87.5	85.4	77.8	64.0	48.4	34.9		
		MCWB	32.2	30.9	42.8	51.8	58.5	64.0	65.7	66.8	60.1	49.6	40.8	29.6		
	2%	DB	28.6	32.2	48.2	64.0	75.9	83.9	82.9	81.2	70.0	56.5	43.0	30.4		
		MCWB	24.8	27.5	38.0	48.5	55.9	62.4	65.0	64.6	56.8	46.5	37.1	25.9		
	5%	DB	23.0	28.6	43.0	59.2	71.3	80.4	79.7	76.4	65.5	51.9	39.2	27.1		
		MCWB	19.8	24.8	34.8	45.8	54.2	61.5	63.9	62.5	54.6	44.0	34.1	23.1		
	10%	DB	17.7	24.1	39.2	54.1	66.7	76.9	76.2	72.7	61.6	47.9	35.2	23.7		
		MCWB	15.1	20.9	32.5	43.0	52.4	60.3	62.5	61.2	52.1	41.2	30.9	20.6		
	0.4%	WB	32.2	31.2	43.6	53.0	61.3	66.6	69.4	68.7	62.0	50.8	41.3	29.9		
		MCDB	36.6	36.2	54.7	68.3	75.8	81.4	83.0	81.4	75.2	60.8	47.9	34.3		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	25.3	27.9	38.7	49.3	58.2	64.8	67.1	65.7	58.2	47.2	37.5	26.1		
		MCDB	28.6	32.2	47.2	62.3	72.5	78.7	79.4	77.0	67.6	56.0	42.9	30.0		
	5%	WB	19.9	24.7	35.5	46.2	55.6	63.2	65.3	63.6	55.5	44.2	34.2	23.2		
		MCDB	23.1	28.7	42.5	57.7	68.7	76.6	75.9	74.4	63.9	51.2	38.8	26.8		
	10%	WB	15.2	20.9	32.5	43.5	52.9	61.4	63.7	61.9	53.0	41.4	31.0	20.6		
		MCDB	17.6	24.0	38.7	53.3	65.1	74.3	73.3	71.4	60.4	47.4	35.1	23.8		
(35) Mean Daily Temperature Range	5% DB	MDBR	14.9	17.2	20.0	21.0	21.5	21.7	18.5	19.7	19.7	17.7	14.7	14.2		
		MCDBR	17.3	19.1	24.6	30.1	29.9	28.1	24.8	27.3	27.4	23.5	17.1	15.9		
		MCWBR	14.7	15.2	16.2	17.3	15.3	11.6	11.0	12.9	15.7	14.8	12.1	13.1		
		MCDBR	16.9	18.8	23.2	27.9	25.9	23.9	20.8	23.8	24.7	22.7	16.8	15.8		
	5% WB	MCWBR	14.5	15.1	16.0	16.9	14.2	10.8	10.4	12.1	14.9	14.5	12.2	13.1		
		MCWBR	14.5	15.1	16.0	16.9	14.2	10.8	10.4	12.1	14.9	14.5	12.2	13.1		
(40) Clear-Sky Solar Irradiance	taub		0.269	0.305	0.355	0.400	0.406	0.398	0.410	0.395	0.357	0.338	0.308	0.273		
	taud		2.087	2.047	2.013	2.131	2.188	2.310	2.298	2.311	2.356	2.279	2.134	2.043		
	Ebn at Noon		213	243	255	261	266	270	264	261	257	232	193	181		
	Edh at Noon		22	32	42	43	43	38	38	36	30	26	21	19		
(44) All-Sky Solar Radiation	RadAvg		318	644	1107	1427	1680	1918	1736	1457	1025	567	282	205		
	RadStd		25	47	91	79	166	160	109	114	95	52	33	22		

Historical Trends

		Heating		Cooling			Degree-Days				
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
(46)	Station Only	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(46) N/A
(47)	Regional (11 neighbors)	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(47) N/A

Nomenclature: See separate page